

Prescribed finite-window reaction-time modality by conserved-budget support design

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Internal working draft—not a submission artifact. Can a fixed amount of spatial reactivity be arranged to prescribe the finite-window topology of an encounter-reaction-time density? We answer this existence question for a Doi encounter process [1]. For every fixed finite dimension $d \geq 2$, prescribed fixed finite integer m , and compact family of interior catalyst weights $w_j \geq w_* > 0$, an m -dependent support design of ordered narrow slabs along a monotone exposure trajectory generates exactly m nondegenerate maxima and $m - 1$ nondegenerate minima on a declared compact positive-time window. The construction first fixes a sufficiently small noise-and-width scale and only then takes a sufficiently small positive installed reactivity budget. Exhaustiveness follows from a global zero bound for the separated Gaussian clocks and a whole-window complement certificate; compact-positive-time mixed-jet convergence transfers the strict topology to the killed process. The result is existential and deliberately nonuniform: its geometry and admissible scales may depend on d and m , and its small-noise contact factor is asymptotically saturated on the design window. It gives no useful budget threshold, event-mass floor, or topology outside that window. A practical two-dimensional allocation-control realization is therefore a separate numerical question. We state a reproducible validation design based on fixed same-budget controls, interval-certified finite-volume propagation, and an independently powered off-lattice event-law comparison, without promoting calculations that do not yet meet that standard.

I. INTRODUCTION

First-passage and reaction-time densities can be multimodal when distinct trajectory families reach a reactive set on separated time scales [2, 3]. Continuum trap placement can create multiple capture-time peaks [4], while two-particle encounter laws in physical dimensions two and three and exact lattice models exhibit related multirate mechanisms [5–7]. Existing theory also describes spatially heterogeneous absorption, fixed-total-reactivity flux optimization, and encounter-dependent reactions [8–11]. The question addressed here is narrower: can the finite-window *timing topology* of a two-particle reaction be prescribed by support design while the installed amount of a static reactivity resource remains fixed?

The fact that a univariate mixture of m Gaussian components has at most m modes is classical in scale-space and Gaussian-mixture theory [12, 13]. We do not claim that upper bound as new. What is needed here is a margin-bearing construction inside a killed two-particle process: all $2m - 1$ stationary points are made simple and located, the whole declared time-window complement is controlled after multiplication by the encounter factor, and the resulting signature is transferred to positive Doi budget.

For each prescribed fixed finite m , the construction chooses an m -dependent family of catalyst supports and keeps its transport, contact geometry, initial law, and total installed centre-space reactivity fixed over all admissible allocations. Every interior allocation in that constructed family has the same alternating signature of m

maxima and $m - 1$ minima. This is a support-geometry construction under a conserved budget, not a proof that one fixed support family selects different mode counts by reallocation. A weak-reactivity mixed-jet estimate transfers the strict compact-window topology from free exposure to the positive-budget Doi law. These are existence results, not certificates for a chosen finite budget or discretization.

The finite-state and finite-grid predecessor developed Green-function reduction, fixed-budget response identities, finite folds, and reduced-clock screening. Those ingredients are treated here as foundation rather than new headlines; a public identifier and detailed overlap statement must be added before submission. Recent universal first-passage response theory concerns the mean first-passage time under rare perturbation activation [14]; the mixed jet below is instead a model-specific statement for the full transient reaction-time density under a static conserved spatial allocation.

A physical two-dimensional control claim requires a different evidence chain: unchanged one-, two-, and three-mode allocation vectors, complete root exclusion on one common window, mesh/parity/alignment/box convergence, and an independent event-law calculation using the same basin cuts and time windows. The final section records that validation design and the claims it can and cannot support.

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II. CONSERVED-REACTIVITY ENCOUNTER MODEL

A. Budget and killed law

Let $q(x, t)$ be the unreacted configuration density of two mobile particles, with conservative generator $\mathcal{L}$. Write r for their relative coordinate and c for a declared encounter-centre coordinate. For a fixed contact profile $\chi_a(r)$ and normalized nonnegative centre-space profiles $\phi_j(c)$, define

$$\begin{aligned} \kappa_{\mathbf{w}}(c) &= B \sum_{j=1}^J w_j \phi_j(c), & w_j &\geq 0, \\ \sum_j w_j &= 1, & \int \phi_j(c) dc &= 1, \\ K_{B, \mathbf{w}}(x) &= \chi_a(r) \kappa_{\mathbf{w}}(c). \end{aligned} \quad (1)$$

Every admissible allocation has the same installed centre-space resource,

$$\int \kappa_{\mathbf{w}}(c) dc = B. \quad (2)$$

This is neither an integral over the full two-particle configuration space nor a count of reactive grid cells. The killed evolution and reaction-time density are

$$\begin{aligned} \partial_t q_{B, \mathbf{w}} &= (\mathcal{L} - K_{B, \mathbf{w}}) q_{B, \mathbf{w}}, & q_{B, \mathbf{w}}(0) &= q_0, \\ f_{B, \mathbf{w}}(t) &= \langle K_{B, \mathbf{w}}, q_{B, \mathbf{w}}(t) \rangle, & -S'_{B, \mathbf{w}}(t) &= f_{B, \mathbf{w}}(t), \end{aligned} \quad (3)$$

where $S_{B, \mathbf{w}}(t) = \int q_{B, \mathbf{w}}(x, t) dx$. Equation (3) is both the physical mass balance and a mandatory numerical invariant. This volume-reactivity killing representation is part of the established Doi/area-reactivity and Feynman–Kac framework [15–17]; the novelty claim does not attach to Eq. (3) itself.

B. Physical two-dimensional quotient and natural-decay realization

The finite-parameter branch below targets physical dimension two. Translation invariance in the common transverse coordinate gives the exact quotient

$$\Omega_\infty = \mathbb{R}_z \times \mathbb{R}_{r_\parallel} \times \mathbb{T}_W, \quad x = (z, r_\parallel, r_\perp), \quad (4)$$

where the last coordinate is periodic with period W . For two equal-diffusivity particles the free forward operator is

$$\begin{aligned} \mathcal{L}q &= -\nabla \cdot (bq) + \nabla \cdot (D \nabla q), \\ D &= \text{diag}(D/2, 2D, 2D), & b &= (-\gamma(z - \bar{z}), -\gamma r_\parallel, 0). \end{aligned} \quad (5)$$

There is no reflecting face in either unbounded coordinate. Natural decay means the weighted-space realization below, rather than a pointwise boundary condition at infinity.

For the physical four-slab family, specialize the generic field in Eq. (1) to

$$V_{\mathbf{w}}(z, r) = \frac{1}{W} \mathbf{1}_{\{|r|_{\text{mi}} < a\}} \sum_{j=1}^J w_j \phi_j(z), \quad K_{B, \mathbf{w}} = B V_{\mathbf{w}}, \quad (6)$$

where $0 < a < W/2$, the bounded nonnegative profiles obey $\int_{\mathbb{R}} \phi_j(z) dz = 1$, and $\mathbf{w}$ is a real simplex control. The factor W^{-1} accounts for the omitted common transverse coordinate and preserves the centre-space budget in Eq. (2). The sharp disk contact is retained; replacing it by a smoothed profile would define a different model.

The normalized reversible density is

$$\begin{aligned} \pi(x) &= Z_\pi^{-1} \exp \left[-\frac{\gamma(z - \bar{z})^2}{D} - \frac{\gamma r_\parallel^2}{4D} \right], \\ Z_\pi &= \frac{2\pi DW}{\gamma}, & D \nabla \log \pi &= b, \end{aligned} \quad (7)$$

with the uniform factor on $\mathbb{T}_W$ included in Z_π . Put

$$\begin{aligned} \mathcal{H}_\pi &= L^2(\Omega_\infty, \pi dx), \\ X_\pi &= L^2(\Omega_\infty, \pi^{-1} dx), & Uu &= \pi u. \end{aligned} \quad (8)$$

The map $U : \mathcal{H}_\pi \rightarrow X_\pi$ is unitary. Let $\mathcal{C} = C_c^\infty(\mathbb{R}^2) \otimes C^\infty(\mathbb{T}_W)$ and let $\mathcal{V}$ be its closure in the norm

$$\|u\|_{\mathcal{V}}^2 = \|u\|_{\mathcal{H}_\pi}^2 + \int_{\Omega_\infty} (\nabla u)^\top D \nabla u \pi dx. \quad (9)$$

For fixed $B \geq 0$ and $\mathbf{w}$, define

$$\mathbf{a}_{B, \mathbf{w}}(u, v) = \int_{\Omega_\infty} (\nabla u)^\top D \nabla v \pi dx + B \int_{\Omega_\infty} V_{\mathbf{w}} uv \pi dx. \quad (10)$$

Proposition 1 (physical natural-decay realization).— For the data above, $\mathbf{a}_{B, \mathbf{w}}$ is a densely defined, closed, symmetric, nonnegative Dirichlet form on the same domain $\mathcal{V}$ as the free form. In particular,

$$\|u\|_{\mathcal{V}}^2 \leq \|u\|_{\mathcal{H}_\pi}^2 + \mathbf{a}_{B, \mathbf{w}}(u, u) \leq (1 + B \|V_{\mathbf{w}}\|_\infty) \|u\|_{\mathcal{V}}^2. \quad (11)$$

Its unique nonnegative self-adjoint operator $\mathbf{H}_{B, \mathbf{w}}$ generates a symmetric analytic sub-Markov contraction semigroup, and

$$A_{B, \mathbf{w}} = U(-\mathbf{H}_{B, \mathbf{w}})U^{-1} \quad (12)$$

is the form-associated natural-decay realization of $\mathcal{L} - BM_{V_w}$. Its action agrees with the displayed algebraic generator on $\mathcal{C}$; no essential-self-adjointness assertion for the minimal core operator is required.

Let $q_0 \in X_\pi$ be nonnegative with unit mass and put $u_0 = q_0/\pi$. Define the budget-normalized observable

$$F_{B,w}(t) = \langle V_w, q_{B,w}(t) \rangle = B^{-1} f_{B,w}(t) \quad (B > 0). \quad (13)$$

For $r = 0, 1, 2$ and $t \in [\tau, T] \subset (0, \infty)$,

$$\begin{aligned} F_{B,w}^{(r)}(t) &= (-1)^r \langle V_w, \mathbf{H}_{B,w}^r e^{-tH_{B,w}} u_0 \rangle_{\mathcal{H}_\pi}, \\ \sup_{t \in [\tau, T]} |F_{B,w}^{(r)}(t)| &\leq \|V_w\|_{\mathcal{H}_\pi} C_r(\tau) \|u_0\|_{\mathcal{H}_\pi}, \\ C_0(\tau) &= 1, \quad C_r(\tau) = \left(\frac{r}{e\tau}\right)^r \quad (r = 1, 2). \end{aligned} \quad (14)$$

Moreover, $q_{B,w}(t) \geq 0$, $F_{B,w}(t) \geq 0$, and

$$\int_{\Omega_\infty} q_{B,w}(t) dx + B \int_0^t F_{B,w}(s) ds = 1. \quad (15)$$

The Gaussian integrals and direct differentiation give Eq. (7). Bounded nonnegative multiplication preserves the closed free form, its domain, and its Dirichlet property. The representation and spectral theorems then give Eqs. (12) and (14); Gaussian cutoffs allow the weak evolution to be tested against the constant function and yield Eq. (15). Full details and the bounded-box companion statement are given in Secs. S1 and S3 of the Supplemental Material.

This proposition identifies the unbounded physical target and its positive-time observable calculus. The reflecting boxes used below are approximants, not boundary conditions of that target. For the twelve formula-defined fixed-box dyadic families, a separate ideal composition now establishes the reconstructed-resolvent, Mosco, and positive-time conclusions from its source-bound premises. That closure concerns only the ideal formula-defined refinement tails. Equality with the production geometry at level $n = 0$ is not a correlated containment receipt: it does not show that one common ideal mass, flux, gauge, map, and killing member lies inside the saved production intervals, and it does not close the production gauge/evaluator bridge. Project-level and production-level complete C1 therefore remain false. Computable spatial and box-truncation errors for time orders $r = 0, 1, 2$, together with a margin-bearing transfer of the complete stationary-point signature to Ω_∞ , also remain open. Proposition 1 licenses no production finite-volume-to-continuum topology claim.

III. ANALYTICAL RESULTS

A. Compact-positive-time weak-reactivity transfer

Choose an interior reference allocation $\bar{w}$ and a fixed matrix $P \in \mathbb{R}^{J \times (J-1)}$ whose columns span $\{h : \mathbf{1}^\top h = 0\}$. Write $w(\theta) = \bar{w} + P\theta$ and let $\Theta \subset \mathbb{R}^{J-1}$ be compact with all its weights strictly positive. Write $K_{B,w} = BV_w$ and set

$$F_B(t, \theta) = B^{-1} f_{B,w(\theta)}(t), \quad G(t, \theta) = \langle V_{w(\theta)}, e^{t\mathcal{L}} q_0 \rangle. \quad (16)$$

Proposition 2 (mixed-jet transfer).— Let the free generator, with the declared reflecting/periodic conditions, be similar to a nonpositive self-adjoint generator on either the bounded L^2 space or the reversible unbounded space $X_\pi = L^2(\pi^{-1}dx)$. Assume q_0 belongs to that state space and that $V_{w(\theta)}$ is affine in θ , uniformly bounded as a multiplication operator on a complex control tube around Θ , and belongs to the corresponding observable dual. Then, on $I = [\tau, T] \subset (0, \infty)$, for every fixed time order r and finite control multi-index α ,

$$\sup_{I \times \Theta} |\partial_t^r \partial_\theta^\alpha (F_B - G)| \leq C_{r,\alpha,I,\Theta} B. \quad (17)$$

The constant is finite for the fixed tube and fixed data and is independent of B on a declared bounded interval. The proof, including the exact Duhamel sensitivity hierarchy and the bounded/unbounded analytic-semigroup realizations, is contained in Secs. S2–S3 of the Supplemental working source. The Dyson expansion and time/control Cauchy estimates then preserve any fixed finite collection of strict root-tube, curvature, complement, or projected-rank margins once those objects have been explicitly defined. Only the C^2 time part is needed for Theorem 1 below. The statement is local on a compact positive-time window: it is not uniform on $t = O(B^{-1})$, does not identify a useful numerical value of B , and supplies no event-mass lower bound. The complete exact- m proof, including its global zero count, full-complement certificate, slow-factor stability, and fixed- ε Doi transfer, is given in the Supplemental working source. This document nevertheless remains a working paper, not a submission package, because the finite-parameter two-dimensional validation and publication-integrity gates remain open.

B. Prescribed finite modality from separated exposure clocks

The conserved-budget support construction admits encounter processes with any prescribed *fixed finite* number of reaction-time modes. We state this existence result in an exact quotient of a two-particle Doi process; the construction depends on the prescribed dimension and

mode count and does not assert mode-count switching by weight redistribution on one fixed support.

Fix finite $d \geq 2$, finite $m \geq 1$, a transverse period $W > 0$, a contact radius $0 < a < W/2$, positive scales $\ell_0, \rho > 0$, and a compact interval $I = [\tau, T] \subset (0, \infty)$. On $\mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}$, consider the independent midpoint and relative processes

$$\begin{aligned} dZ_t &= -\gamma(Z_t - \bar{z}) dt + \varepsilon \sqrt{D_0} dW_t, \\ dR_{\parallel,t} &= -\gamma R_{\parallel,t} dt + 2\varepsilon \sqrt{D_0} dW_{\parallel,t}, \\ dR_{\perp,t} &= 2\varepsilon \sqrt{D_0} dW_{\perp,t} \pmod{W}. \end{aligned} \quad (18)$$

The mutually independent initial laws satisfy

$$\begin{aligned} Z_0 &\sim N(z_0, \varepsilon^2 D_0 / (2\gamma)), & D_0, \gamma, u_0 &> 0, & z_0 &\neq \bar{z}, \\ R_{\parallel,0} &\sim N(r_{\parallel,0}, \varepsilon^2 u_0^2), & u_0^2 &< 4D_0/\gamma, & \Sigma_{\perp,0} &\succ 0. \\ R_{\perp,0} &\sim \text{WN}(r_{\perp,0}, \varepsilon^2 \Sigma_{\perp,0}), \end{aligned} \quad (19)$$

Here WN is the wrapped Gaussian on $\mathbb{T}_W^{d-1}$, $\Sigma_{\perp,0}$ is a positive-definite $(d-1) \times (d-1)$ covariance matrix, and $|r|_{\text{mi}}$ is the Euclidean norm of the longitudinal relative coordinate together with the shortest transverse representatives modulo W .

All driving Brownian motions are also independent. Thus $\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}$ is strictly monotone and $\text{Var } Z_t = \varepsilon^2 D_0 / (2\gamma)$. Under the minimum-image convention, assume the deterministic relative trajectory $r_*(t) = (r_{\parallel,0}e^{-\gamma t}, r_{\perp,0})$ obeys $\sup_{t \in I} |r_*(t)|_{\text{mi}} \leq a - \eta$ for some $\eta > 0$.

Choose target times

$$\begin{aligned} \tau < t_1 < \dots < t_m < T, & \quad c_j = x(t_j), \\ x(t) &= \frac{\text{sgn}(\mu')\mu(t)}{\ell_0}, \end{aligned} \quad (20)$$

so that $x'(t)$ is bounded away from zero. At the corresponding physical centres, place normalized Gaussian catalyst slabs of width $\varepsilon\rho$. Their nonnegative weights belong to a compact subset of the simplex,

$$\mathcal{W}_{w_*} = \left\{ \mathbf{w} : \sum_{\substack{j=1 \\ w_j \geq w_* > 0}}^m w_j = 1 \right\}, \quad 0 < w_* \leq \frac{1}{m}, \quad (21)$$

and define

$$\begin{aligned} \phi_{j,\varepsilon}(z) &= \frac{\exp[-(z - \mu(t_j))^2 / (2\varepsilon^2 \rho^2)]}{\sqrt{2\pi} \varepsilon \rho}, \\ K_{B,\mathbf{w},\varepsilon}(Z, R) &= \mathbf{1}_{\{|R|_{\text{mi}} < a\}} \frac{B}{W^{d-1}} \sum_{j=1}^m w_j \phi_{j,\varepsilon}(Z). \end{aligned} \quad (22)$$

The slab normalization makes B the same installed centre-space budget for every allocation. Set $K_{B,\mathbf{w},\varepsilon} =$

$BV_{\mathbf{w},\varepsilon}$, and let $T_0(t)$ and q_0 denote the unkilld semi-group and initial law. Then

$$\begin{aligned} G_{\varepsilon,\mathbf{w}}(t) &= B^{-1} \mathbb{E}_0[K_{B,\mathbf{w},\varepsilon}(Z_t, R_t)] = \langle V_{\mathbf{w},\varepsilon}, T_0(t)q_0 \rangle, \\ f_{B,\varepsilon,\mathbf{w}}(t) &= \mathbb{E}_0 \left[K_{B,\mathbf{w},\varepsilon}(Z_t, R_t) e^{-\int_0^t K_{B,\mathbf{w},\varepsilon}(Z_s, R_s) ds} \right]. \end{aligned} \quad (23)$$

The expectation $\mathbb{E}_0$ is with respect to the unkilld process started from Eq. (19). Independence gives the exact factorization

$$\begin{aligned} G_{\varepsilon,\mathbf{w}}(t) &= A_{\varepsilon}(t) H_{\sigma,\mathbf{w}}(x(t)), \\ A_{\varepsilon}(t) &= \frac{c_{d,\varepsilon}(t)}{W^{d-1} \sqrt{2\pi} \varepsilon \sqrt{D_0 / (2\gamma) + \rho^2}}, \end{aligned} \quad (24)$$

where $c_{d,\varepsilon}(t) = \mathbb{P}_0(|R_t|_{\text{mi}} < a)$ and $H_{\sigma,\mathbf{w}}$ is defined below.

Theorem 1 (exact prescribed finite modality).— Fix finite $d \geq 2$, finite $m \geq 1$, the data above, and the compact window I . There is $\varepsilon_0 > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$ and every $\mathbf{w} \in \mathcal{W}_{w_*}$, the unit-budget free-exposure clock has exactly m nondegenerate maxima and $m-1$ nondegenerate minima on I , ordered alternately, and no stationary endpoint. For each fixed such ε , there is $B_0(\varepsilon) > 0$, uniform over $\mathcal{W}_{w_*}$, such that the budget-rescaled Doi reaction-time density $f_{B,\varepsilon,\mathbf{w}}/B$ has the same complete stationary signature for every $0 < B < B_0(\varepsilon)$. Multiplication by B leaves the stationary points unchanged, so the conclusion also holds for the physical density $f_{B,\varepsilon,\mathbf{w}}$.

The mechanism can be seen without invoking a normal form. Gaussian convolution reduces the longitudinal part of the unit-budget exposure to

$$\begin{aligned} H_{\sigma,\mathbf{w}}(x) &= \sum_{j=1}^m w_j \exp \left[-\frac{(x - c_j)^2}{2\sigma^2} \right], \\ \sigma &= \frac{\varepsilon}{\ell_0} \sqrt{\frac{D_0}{2\gamma} + \rho^2}. \end{aligned} \quad (25)$$

Writing $q_j(x) = w_j \exp[-(x - c_j)^2 / (2\sigma^2)]$, $\pi_j(x) = q_j(x) / H_{\sigma,\mathbf{w}}(x)$, and $L = \partial_x \log H$, let $\bar{c}(x)$ and $\text{Var}_{\pi(x)}(c)$ denote the mean and variance of the centres under the finite probability vector $\boldsymbol{\pi}(x)$. Then

$$L(x) = \frac{\bar{c}(x) - x}{\sigma^2}, \quad L'(x) = \frac{\text{Var}_{\pi(x)}(c)}{\sigma^4} - \frac{1}{\sigma^2}. \quad (26)$$

The univariate upper bound of m modes for an m -component Gaussian mixture is classical [12, 13]. Here a direct affine-exponential proof is retained because the Doi transfer needs all critical zeros counted with multiplicity together with uniform root-tube, complement, curvature, and endpoint margins.

For the adjacent pair $c_j < c_{j+1}$, the weighted crossover is

$$s_j = \frac{c_j + c_{j+1}}{2} + \frac{\sigma^2}{c_{j+1} - c_j} \log \frac{w_j}{w_{j+1}}. \quad (27)$$

The adjacent ratio obeys $q_{j+1}/q_j = 1/9$ at $s_j - \sigma^2 \log(9)/(c_{j+1} - c_j)$ and $q_{j+1}/q_j = 9$ at the corresponding plus edge. Thus both adjacent components remain uniformly present throughout the crossover, while all nonadjacent components are exponentially small. Equation (26) then yields one simple minimum at

$$r_j = s_j + O(\sigma^4), \quad L'(r_j) = \frac{(c_{j+1} - c_j)^2}{4\sigma^4} + O(\sigma^{-2}). \quad (28)$$

Each centre similarly contributes one simple maximum at $p_j = c_j + O(\exp[-q/\sigma^2])$, with $L'(p_j) = -\sigma^{-2} + o(\sigma^{-2})$. These constructions give $2m - 1$ simple stationary points. They are exhaustive: after multiplication by a positive Gaussian factor, H' is an exponential polynomial

$$\sum_{j=1}^m (a_j + b_j x) e^{\lambda_j x}, \quad \lambda_1 < \dots < \lambda_m, \quad (29)$$

which has at most $2m - 1$ real zeros counted with multiplicity. The extreme affine-exponential terms dominate in the two tails, so all zeros lie in a compact interval and real analyticity makes their number finite. A generalized Rolle induction then proves the bound by differentiating twice to remove the lowest-exponent affine term.

The remaining relative-coordinate contact probability is a positive slow factor. Uniform bounds on its first two logarithmic time derivatives, peak boxes of width $O(\sigma^2)$, valley boxes of width $O(\sigma^4)$, and a posterior-sector sign certificate exclude stationary points on the full box complement. Consequently the slow factor moves maxima by $O(\sigma^2)$ and minima by $O(\sigma^4)$ without changing their number or type. Finally, on every fixed positive ε , compact-positive-time mixed-jet convergence

$$\sup_{\mathbf{w} \in \mathcal{W}_{w_*}} \left\| \frac{f_{B, \varepsilon, \mathbf{w}}}{B} - G_{\varepsilon, \mathbf{w}} \right\|_{C^2(I)} \rightarrow 0 \quad (B \downarrow 0) \quad (30)$$

preserves the disjoint root tubes, signed curvature margins, complement margin, and endpoint slopes.

The quantifier order is essential: the geometry and admissible scales may depend on fixed (d, m) ; one first fixes a sufficiently small $\varepsilon > 0$ and only then chooses $B < B_0(\varepsilon)$. The theorem does not provide a useful uniform lower bound for B_0 , a uniform-in- d or uniform-in- m construction, an event-mass floor, or a stationary-point count outside I ; it also uses a whole-window contact-interior condition whose small-noise limit is asymptotically saturated contact. Nontrivial contact, observability at a common positive budget, and finite-parameter robustness remain separate numerical validation gates.

IV. FINITE-PARAMETER EVIDENCE BOUNDARY

The theorem above proves existence inside an explicitly constructed, parameter-dependent family. It does not certify the broad four-slab finite-volume family used to seek a practical two-dimensional control. Earlier computations in that family provide design information only: one unchanged allocation at $B = 0.01$ retained five alternating stationary roots on odd meshes $N = 113$ and $N = 129$ in one reflected box and the same finite-volume solver family. The saved floating-point screen searched roots only through $t = 35$, while basin and survival calculations extended to $t = 100$; no post-35 root-exclusion certificate was made. Those calculations neither exclude later extrema nor establish parity, alignment, box, continuum, or independent-method robustness. A later allocation-cusp search did not produce a claimable branch and is not used here.

Three exact-rational, same-budget allocations have instead been fixed for a prospective one-/two-/three-mode test. Their positive-budget values have not been evaluated or read in this working process. No statement here asserts that those controls realize the target topologies, that a finite-budget cusp exists, or that a bounded finite-volume calculation represents an unbounded continuum law.

V. REPRODUCIBLE TWO-DIMENSIONAL VALIDATION DESIGN

Before any positive-budget value may be evaluated or read, the transport, contact radius, initial law, slab supports, installed budget, allocation vectors, observation window, acceptance margins, rate-defined generator, and common-observable rule must be frozen and independently accepted. The deterministic campaign then executes the predeclared 36-row mesh, parity, alignment, and box family without refitting or adaptive repair after a first scientific value is seen.

A. Verified finite-volume propagation

For each prescribed mesh, parity, slab alignment, and box, a rate-defined uniformization calculation must enclose the density, its first two time derivatives, survival, and valley-partitioned masses. A separately replayable interval certificate must include every stationary point with strict curvature sign and exclude all other stationary points on the complete declared window. Mass balance and tail truncation are checked in the same arithmetic envelope; implementation-level provenance and memory-isolation controls belong to the reproducibility record rather than to the physical claim.

The control-free preparation covers all 12 prospectively fixed production configurations. Exact axis par-

titions, free-axis forward/backward rate enclosures, initial marginals, and sparse tensor boxes are stored in a canonical file-backed bundle representing 34,787,462 states. The historical file role `stationary_mass` does not contain a physical cell integral: on axis a it is the ungauged representative quadrature primitive

$$\tilde{\mu}_{a,i} = |C_{a,i}| \exp[-\Phi_a(x_{a,i}^{\text{rep}})].$$

It is neither box-normalized nor equal, in general, to $\int_{C_{a,i}} \pi_a(x) dx$.

A same-core artifact implementation reconstructs every canonical byte. A separate-source implementation checks containment of the partitions, directed rates, ungauged representative primitives, initial marginals, and active tensor components. This is source-level semantic containment, not an independent numerical backend.

Separate authenticated fixed-row sources now enclose the physical axis-cell integrals and reconstruct the formula-defined common edge flux $\kappa_{a,i+1/2}$, the corresponding directed rates, and the factorized density ratio. Their producers and independently written validators use the same pinned `gmpy2`/MPFR backend. Authentication of the executed bytes and runtime therefore supplies provenance, not backend independence. These fixed-row sources do not by themselves give a genuine refinement theorem or one correlated same-member production receipt.

The canonical rate files are joined to native packed payloads and round-tripped byte for byte. Two serialized replays of the initial bundle and a further two-repeat replay of the control-free factorized killing geometry close only their declared file/process boundaries. The latter contains no concrete control, positive budget, tensor killing diagonal, reconstructed multiplier, or full operator. Separate ideal records define twelve genuine dyadic refinement sequences and source-bound symbolic map, cut-layer, and killing-residual estimates, but they do not bind the production member. Consequently no external predecessor commitment or fresh ordered roles 8–10 replay exists. Same-member containment, the production resource gate, F0–F3, complete C1–C3, continuum topology and componentwise root transfer, the physical control claim, release, and submission eligibility all remain held.

The three controls are accepted as one family only if all prescribed configurations retain, respectively, exactly one, two, and three maxima and the associated basin masses remain above the declared numerical resolution floor. A failure of any configuration is reported at that bounded scope; it is not repaired by dropping a mesh, changing a threshold, or replacing a control.

B. Common observables and an independent event law

Only after the complete deterministic family is independently accepted may a predeclared mechanical rule

form common basin cuts and time windows from its envelopes. The off-lattice killed-process calculation then uses unchanged physical inputs and two independent random-number pools: one for prescribed precision decisions and one for the reported estimates. Simultaneous confidence intervals test basin probabilities, survival, and common-window contrasts. Trajectory samples do not certify time derivatives or a continuous-density norm, and repeated quadrature or sparse-matrix evaluation of the same finite-volume generator is not an independent physical method.

C. Continuum claim boundary

A claim of a resolved physical two-dimensional realization requires agreement of all retained root intervals, topology margins, survival values, and basin masses across the full mesh/parity/alignment/box family, followed by the off-lattice comparison. The result would remain finite-window and same-family: three fixed controls do not by themselves establish a cusp, global phase diagram, topology outside the window, or a positive-budget three-dimensional result. Even complete agreement in that deterministic and off-lattice campaign would constitute continuum-consistent evidence only on the declared finite window; it would not certify convergence from a finite-volume scheme to the unbounded continuum. That certification additionally requires the C0–C3 map, same-member, quantitative discretization, and box-truncation chain together with margin-bearing componentwise root transfer.

VI. DISCUSSION

The analytical mechanism is constructive. A monotone midpoint trajectory converts separated catalyst locations into separated exposure clocks; narrow Gaussian convolution creates one peak near each target time; a global zero bound and a whole-window sign certificate make the count exhaustive; and a weak-reactivity mixed-jet estimate preserves the resulting topology in the Doi law. This establishes arbitrary prescribed *fixed finite* modality without appealing to a reduced-clock universality claim.

The restrictions are scientifically consequential. Geometry and smallness scales depend on d and m . The deterministic relative trajectory stays strictly inside the contact set on the designed window, and the free contact probability tends to one there in C^2 as the small-noise scale vanishes; no pathwise Brownian confinement is claimed. The limits are sequential, and observability at a common useful budget is not guaranteed. The next contribution is therefore not another parameter scan. It is the independently reproducible two-dimensional validation described in Sec. V. Until that chain is complete, the theorem and the historical computations remain separate results.

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